

# Random Forests

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# Outline

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# From One Tree to a Forest

- A single Decision Tree is easy to interpret but sensitive to data.
- Slight changes in input can lead to completely different trees.
- **Random Forests** solve this by building many trees and combining their outputs.

**Idea:** Aggregate the wisdom of many trees to reduce overfitting and improve prediction.

**Random Forest** is an ensemble algorithm that builds multiple decision trees and merges their results.

- For classification: output is majority vote across trees.
- For regression: output is average of predictions.
- Each tree is trained on a slightly different dataset.

**Goal:** Reduce variance without increasing bias too much.

Random Forest introduces two levels of randomness:

- ① **Bootstrap Sampling:** Each tree sees a random sample (with replacement) of training data.
- ② **Feature Subsets:** At each node, only a random subset of features is considered for splitting.

This ensures that trees are **decorrelated**, leading to a better ensemble.

# Bagging: Bootstrap Aggregation

Bagging reduces variance by averaging many noisy but unbiased models.

- Suppose we have  $T$  trees:  $f_1, f_2, \dots, f_T$
- Final prediction:  $\hat{f}(x) = \frac{1}{T} \sum_{t=1}^T f_t(x)$
- Each  $f_t$  is trained on a different random subset of data

Bagging works especially well with models like trees that have high variance.

# Why Feature Randomness?

Without random feature selection:

- Many trees may pick the same top feature at the root.
- Ensemble becomes less diverse, reducing effectiveness.

By selecting a random subset of features at each node:

- Trees explore different splits.
- Results are less correlated, improving ensemble quality.

# Advantages of Random Forests

- **High accuracy:** Often among the best-performing models out-of-the-box.
- **Robust to overfitting:** Aggregation reduces variance.
- **Handles missing data:** Fairly resilient to noisy or incomplete features.
- **Non-linear modeling:** No assumptions about data distribution.

Also provides **feature importance scores**, useful for interpretation.



- **Interpretability:** Not as simple to explain as a single tree.
- **Computational cost:** More trees = more memory and time.
- **Not always best:** For small datasets, simpler models may perform better.

Still, Random Forests are a great default choice in most supervised learning problems.

- Random Forest = many Decision Trees + randomization + aggregation.
- Improves performance and stability over single trees.
- Easy to use, powerful, and widely adopted.
- Good next step after learning Decision Trees.

**Next:** Try Random Forests on real datasets and compare them to single-tree results!